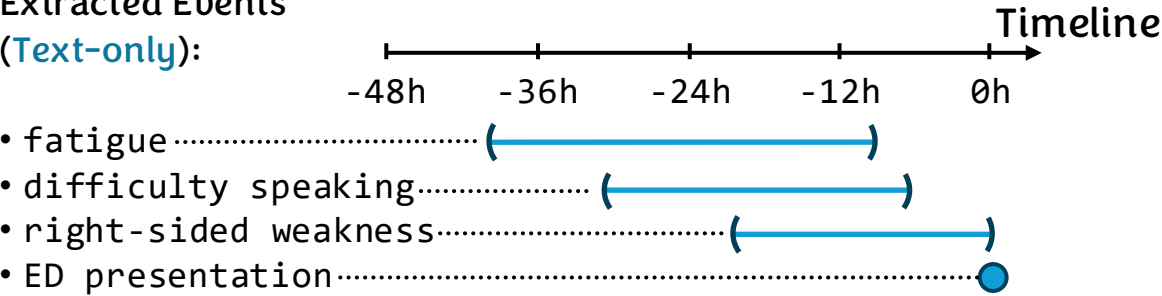


(A) Textual Ambiguity (Text-only timeline reconstruction)

Clinical Note (T)

Patient presented to ED with acute onset right-sided weakness. Reports fatigue over the past day. Family notes difficulty speaking starting earlier in the day.

Extracted Events (Text-only):



Observations:

- ✓ Endpoint (ED presentation) is correct
- ✓ Rough ordering is partially preserved
- ✗ Mid-trajectory events have wide and overlapping uncertainty
- ✗ Cannot distinguish progression timing (fatigue → speech → weakness)

Multimodal alignment using structured EHR anchors

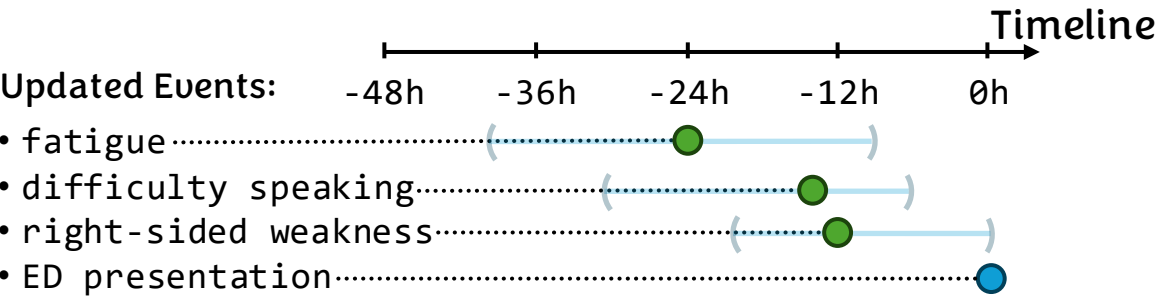
(B) Tabular Calibration (Multimodal alignment)

Structured EHR Rows (R)

Event	Value	Timestamp
WBC Lab	14.2×10 ³ /μL	04/14/2026 04:00
Vitals	Tachycardia (HR 110)	04/16/2026 02:00
...		
Triage Note	Speech deficit	04/16/2026 06:00
ED Arrival	—	04/16/2026 08:00
...		

Structured anchors: (EHR Rows):

- WBC → fatigue..... (-20h)
- Vitals → weakness..... (-6h)
- Triage → speech..... (-2h)
- ED Arrival..... (0h)



Improvements:

- ✓ Events align to clinically meaningful anchors
- ✓ Temporal ordering becomes clearer
- ✓ Mid-trajectory ambiguity is resolved
- ✓ Progression becomes interpretable (fatigue → speech → weakness)

- (—) Text-derived event interval
- (—) Prior wide event interval
- Text-derived event time
- ◆ Structured EHR anchor
- Calibrated event time